

Interaksi Manusia dan Komputer (IMK)

Pengantar IMK



Tujuan Mata Kuliah IMK

Setelah mengikuti mata kuliah ini mahasiswa dapat:

“**Merancang** antarmuka perangkat lunak yang memenuhi **prinsip *User Interface (UI)*** yang benar”

Outline

- Definisi IMK
- Pentingnya desain UI yang baik
- Perkembangan IMK
- Tren UI/UX
- Interaction Framework
- Interaction Style
- Karakteristik GUI dan WebUI

Definisi Interaksi Manusia Komputer

Definisi IMK/HCI

- Human-computer interaction is the **study, planning**, and **design** of how **people and computers work together** so that a person's needs are **satisfied** in the most **effective** way (Galitz, 2007)
- Human-computer interaction is a discipline concerned with the **design, evaluation and implementation of interactive computing systems** for human use and with the study of major phenomena surrounding them (Hewett et al, 1996)

HCI / IMK

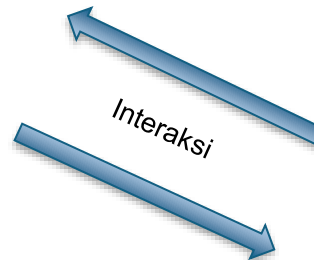
- Pada saat melakukan design antarmuka kita perlu memperhatikan beberapa hal:
 - apa keinginan dan harapan orang terhadap produk yang dibangun,
 - apa batasan dan kemampuan fisik para penggunanya,
 - memperhatikan *user experience* dari pengguna produk, Pertimbangkan apa yang menurut pengguna baik dan buruk dari antarmuka yang dirancang, dan
 - Lakukan selalu test dan retest agar mendapatkan design antarmuka yang sesuai dengan kebutuhan pengguna.
- Perancang juga harus memperhatikan **karakteristik** dan **batasan teknis** dari **perangkat keras** dan **perangkat lunak** komputer

Manusia Vs Komputer



Manusia:

- fleksibel & mampu beradaptasi,
- dapat belajar bagaimana bekerja di lingkungan yang baru



Komputer:

- tidak fleksibel / tidak mampu beradaptasi,
- input harus dalam format yang jelas & output harus didefinisikan sebelumnya,
- tidak dapat belajar,
- dapat didesain ulang



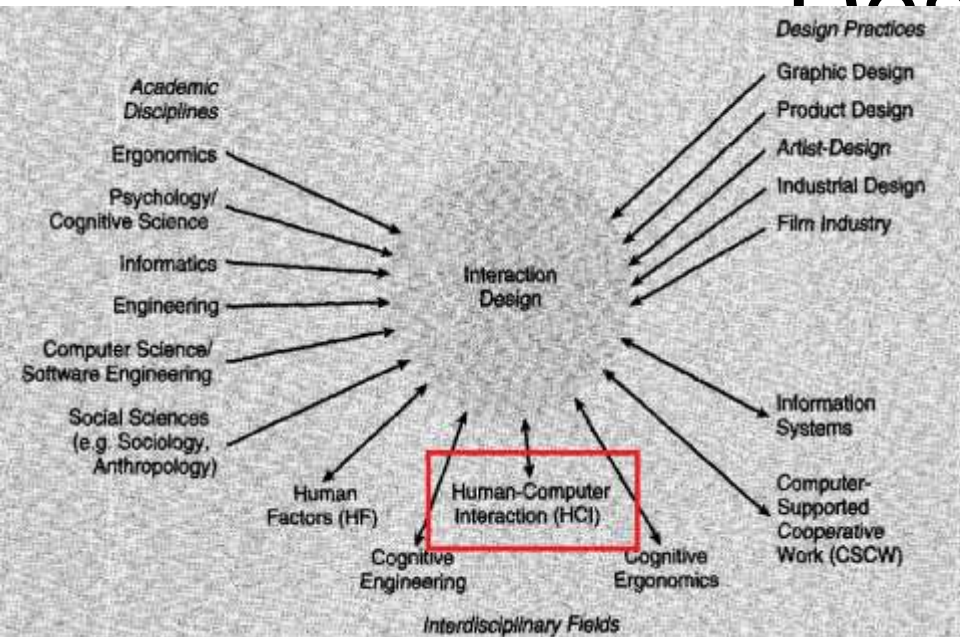
Paradigma Interaksi

- Paradigma interaksi adalah model atau pola interaksi manusia komputer yang mencakup semua aspek interaksi, termasuk fisik, virtual, persepsi, dan kognitif (Heim 2007).
- Paradigma interaksi diidentifikasi oleh Heim(20017) :
 - Large Scale Computing
 - Personal computing
 - Networked computing
 - Mobile Computing
 - Collaborative Environment
 - Virtual Reality
 - Augmented Reality

5 W + H

- Who, what, where, why, and how
- 5 W + H heuristic adalah prosedur yang dapat digunakan untuk mendefinisikan dan menganalisa paradigma interaksi yang ada dan ruang serta mengeksplorasi unsur-unsur dan objek dengan yang berinteraksi pengguna.
- Heuristik itu terdiri dari 3 komponen :
 - What/How : digunakan untuk memahami komponen antarmuka fisik dan virtual.
 - Where/When : Hal ini terkait dengan lingkungan fisik. terlihat perbedaan antara kantor, portabel, sistem dpt dipakai.
 - Who/Why : ini terlihat pada jenis tugas dan keahlian yang dibutuhkan.
- **Petakan 5 W + H terhadap jenis paradigma interaksi menurut Heim**

Hubungan HCI/IMK - Interaction Design



Jenny Preece, Yvonne Rogers, Helen Sharp. 2002.
*Interaction Design_beyond Human-Computer –
Interaction*, J. Wiley & Sons

HCI/IMK merupakan disiplin ilmu yang dibutuhkan dalam merancang sebuah antarmuka sebuah produk dalam hal ini produk *software*

Pentingnya Desain UI yang Baik

User Interface

- Antarmuka/ *user interface* (UI) merupakan bagian dari komputer dan perangkat lunaknya yang dapat dilihat, didengar, disentuh, dan diajak bicara, baik secara langsung maupun dengan proses pemahaman tertentu.
- UI yang baik adalah UI yang tidak disadari, dan UI yang memungkinkan pengguna fokus pada informasi dan task tanpa perlu mengetahui mekanisme untuk menampilkan informasi dan melakukan task tersebut.
- Komponen utamanya:
 - Input
 - Output

Basics of IT applications

Stand-alone application: computer program [input → do something → output]

Complex applications: Many interacting computer programs

Fundamental component:

IT applications take inputs (from sensors, **humans**)

Most IT applications provide outputs (to actuators, **humans**)

→ A “Good” design of the User-Computer Interface is important !



Pentingnya Desain UI yang Baik

- Banyak sistem dengan fungsionalitas yang baik tapi tidak efisien, membingungkan, dan tidak berguna karena desain UI yang buruk
- Antarmuka yang baik merupakan jendela untuk melihat kemampuan sistem serta jembatan bagi kemampuan perangkat lunak
- Desain yang buruk akan membingungkan, tidak efisien, bahkan menyebabkan frustrasi

Desain UI yang Baik perlu memenuhi

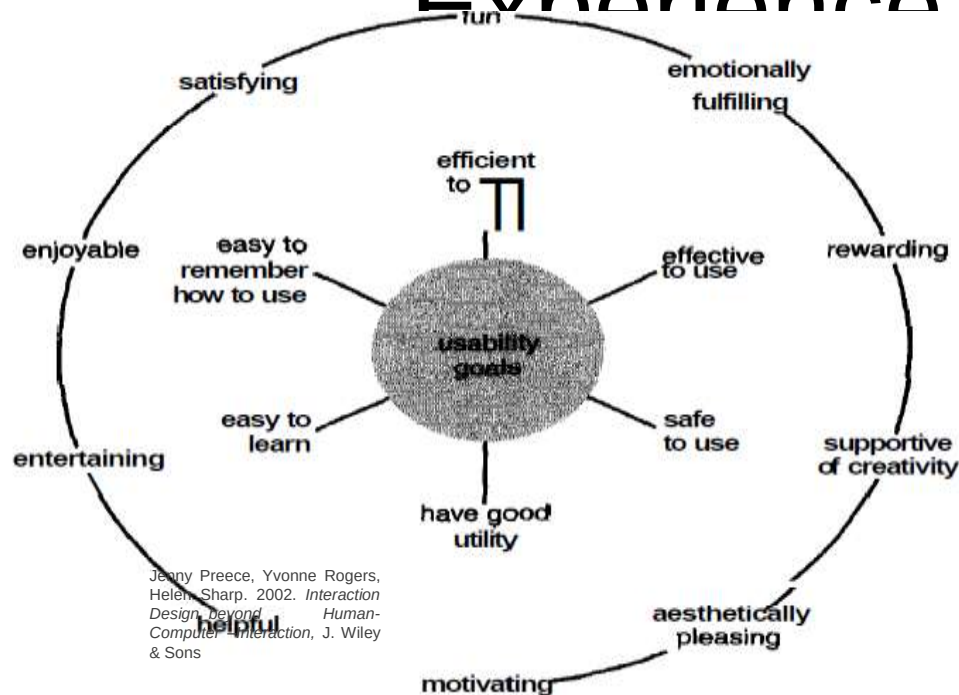
- **Usability goals**

- effective to use (effectiveness)
- efficient to use (efficiency)
- safe to use (safety)
- have good utility (utility)
- easy to learn (learnability)
- easy to remember how to use (memorability)

- **User Experience goals**

- Satisfying -
- Enjoyable
- Fun
 - Entertaining
- Helpful
 - Motivating
- Aesthetically pleasing - Supportive of creativity
- Rewarding -
- Emotionally fulfilling

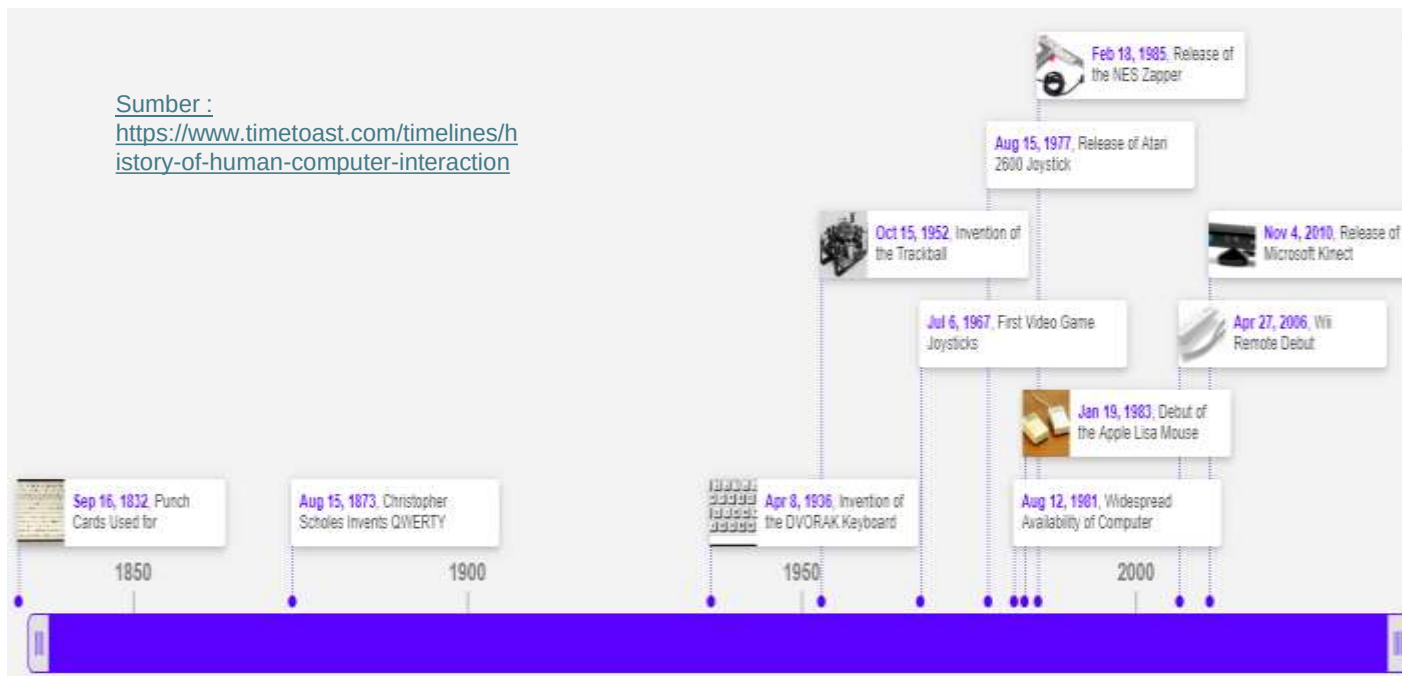
Usability dan User Experience goals



Jenny Preece, Yvonne Rogers,
Helen Sharp. 2002. *Interaction
Design, Beyond Human-
Computer Interaction*, J. Wiley
& Sons

Perkembangan IMK

Sejarah perkembangan IMK



Tren Interaksi Manusia Komputer

- Smart Restaurant Table
 - <https://www.youtube.com/watch?v=UZPwqImfBf0>
 - <https://itrestaurant.net/table>
- Amazing Interface & Holograms
 - <https://www.youtube.com/watch?v=uaLWylmrpCI>
 - <https://collegeinfo geek.com/leap-gesture-computing/>



Sumber : <https://itrestaurant.net/table>



Sumber : <https://collegeinfo geek.com/leap-gesture-computing/>



Sumber :
<http://sparrowinteractiondesign.blogspot.com/2015/04/cicret-bracelet-future-hci-technologies.html>

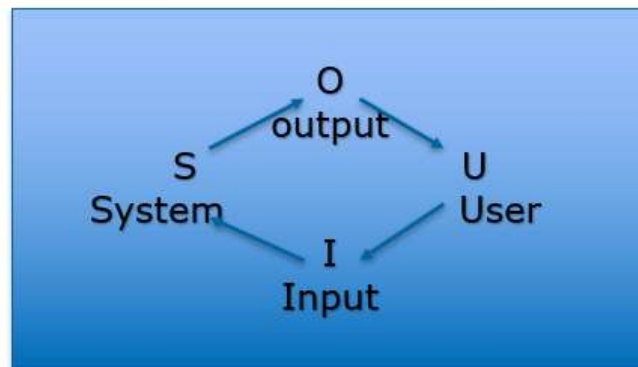


Sumber :
<https://www.canterbury.ac.nz/engineering/schools/csse/research/hci/>

Interaction Framework

Interaction Framework

- Manusia berinteraksi dengan komputer tentunya sangat kompleks dan tidak mudah.



Interaction Framework

Interaction Style

Interaction Style

- An interaction style is simply the method, or methods, by which the user and a computer system communicate with one another
- Type of Interaction Style :
 - Command line
 - Menu-Based Interface
 - Form Fill in
 - Direct Manipulation
 - Anthropomorphic/Metaphors
 - Web Navigation
 - 3D Environment
 - Zoomable interface
 - Natural Language

Interaction Style

➤ Command line :

- Merupakan *interaction style* paling awal dari teknologi computer
- Powerfull for Advanced user
- Complex Syntax
- “Low error tolerant” sehingga bisa menyebabkan user frustrasi

```

DB2 CLP - DB2 - C:\PROGRA~1\IBM\SQLLIB\bin\db2setcp.bat DB2SETCP.BAT DB2.DOC
EDIT          PREP/PRECOMPILE      UPDATE ALERT CFG
EXPORT        PHRASE HISTORY/LOGFILE UPDATE COMMAND OPTIONS
FORCE APPLICATION PUT ROUTINE      UPDATE CONTACT
GET/UPDATE ADMIN CFG  QUERY CLIENT    UPDATE CONTACTGROUP
GET ALERT CFG        QUIESCE DATABASE  UPDATE NOTIFICATION LIST
GET AUTHORIZATIONS   QUIESCE INSTANCE  UPDATE HISTORY
GET/UPDATE CLI CFG   QUIESCE TABLESPACE  UPDATE LDAP MODE
GET CONNECTION STATE  QUIT          UPDATE MONITOR SWITCHES
GET CONTACTGROUP     REBIND
GET CONTACTGROUPS    RECONCILE

Note: Some commands are operating system specific and may not be available.

For further help:
? db2-command - help for specified command
? OPTIONS - help for all command options
? HELP - help for reading help screens

The preceding three options can be run as db2 <option> from an OS prompt.

!db2ic - DB2 Information Center (Windows only)
This command can also be run as db2ic from an OS prompt.
db2 -> describe avarsdmanager
SQLERROR An unexpected token "avarsdmanager" was found following "DESCRIBE".
Expected tokens may include: "OUTPUT", SQLSTATE=42601
db2 -> ? describe

```

Interaction Style

Menu Selection

- Trend UI saat ini di berbagai macam Aplikasi.
- Utilize a person with much stronger Recognition.
- Membantu bagi *infrequent user*.
- Label Menu harus dimengerti dan memiliki arti yang jelas



Interaction Style

- Form Fill in
 - Sangat berguna untuk mengambil Informasi
 - Membutuhkan *typing skill* yang tinggi
 - Familiar dengan *paper form* di dunia nyata



Interaction Style

► Direct Manipulation

- Memungkinkan User berinteraksi secara langsung bahkan memanipulasi objek yang tampil di layar



Interaction Style

- **Anthropomorphic**
 - Sebuah antarmuka anthropomorphic mencoba untuk berinteraksi dengan orang-orang dengan cara yang sama seperti orang-orang berinteraksi satu sama lain. Ini mencakup dialog bahasa alami lisan, gerakan tangan, ekspresi wajah dan gerakan mata



Interaction Style

STYLE	ADVANTAGES	DISADVANTAGES
Command Line	Powerfull, Flexible, Appeals to expert user, Conserves screen space	Commands must be memorized, requires big learning effort, intolerant of typing errors
Menu Selection	Utilizes recognition memory, reduces interaction complexity, aids decision making process, minimize typing, aids casual users	Consumes screen space, may create complex menu hierarchies, may slow knowledgeable users
Form Fill in	Familiar format, simplifies information entry, requires minimal training	Consumes screen space, requires carefull and efficient design, does not prevent typing errors
Direct Manipulation	Faster Learning, easier remembering, exploits visual/spatial cues, easy error recovery, provides context, immediate feedback	Greater design complexity, window manipulation requirements, requires icon recognition, inefficient for touch typist, increased chance for screen clutter
Anthropomorphic	Natural	Difficult to implement

Karakteristik GUI dan Web UI



Text Based VS GUI VS Web UI

TDX95210 THE CAR RENTAL COMPANY 18/11/76 18:25

NAME TEL RD

PUD RD C RT MPD

ENTRY ERROR:00465628996Q.997

Command==>

Text Based



GUI



Web UI

GUI VS WEB UI



KARAKTERISTIK SISTEM DIRECT MANIPULATION

Merupakan potret langsung dari kehidupan nyata

Object dan action selalu terlihat

Rapid and incremental Action with visible display of results

Incremental action are easily reversible

Graphics User Interface (GUI)

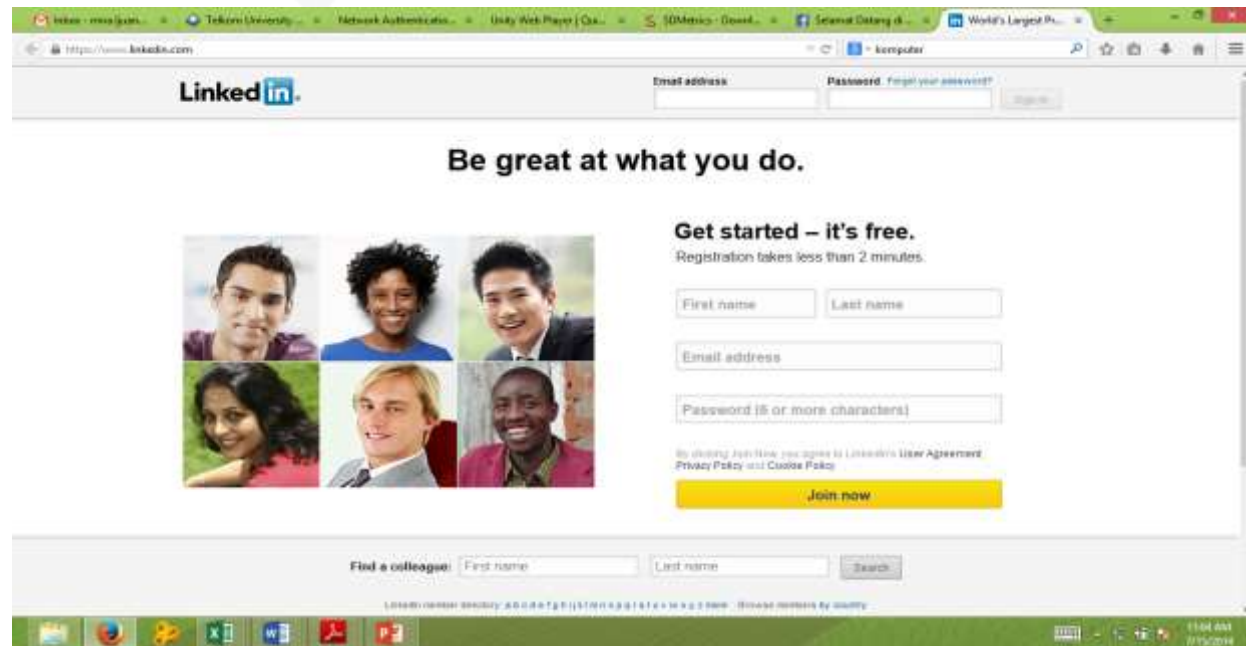
STYLE	ADVANTAGES	DISADVANTAGES
Direct Manipulation	Faster Learning, easier remembering, exploits visual/spatial cues, easy error recovery, provides context, immediate feedback, more attractive, easily augmented with text display, low typing req.	Greater design complexity, window manipulation requirements, requires icon recognition, inefficient for touch typist, increased chance for screen clutter, note always familiat, human comprehension limitation, may consume more screen space, hardware limitation.

Characteristics Of GUI

- Sophisticated Visual Presentation
- Pick and click interaction
- A restricted set of interface solution
- Visualization
- Object orientation
- Extensive use of a person's recognition memory
- Concurrent performance of function

Web User Interface

- Hal yang paling mendasar dalam Desain antarmuka Web itu adalah pada desain navigasi dan penyajian informasi. Dalam Web yang paling penting adalah Konten bukan Data
- Web Page VS Web Application
- More user Control



- Cari tahu lebih banyak perbedaan GUI dan Web dari aspek Content, Interaksi

Table 2.2: GUI versus Web Design

	GUI	WEB
Devices	User hardware variations limited. User hardware characteristics well defined. Screens appear exactly as specified.	User hardware variations enormous. Screen appearance influenced by hardware being used.
User Focus	Data and applications.	Information and navigation.
Data/Information	Typically created and used by known and trusted sources. Properties generally known. Typically placed into system by users or known people and organizations. Typically organized in a meaningful fashion. A notion of private and shared data exists.	Full of unknown content. Source not always trusted. Often not placed onto the Web by users or known people and organizations. Highly variable organization. Privacy often suspect.
User Tasks	Install, configure, personalize, start, use, and upgrade programs. Open, use, and close data files. Fairly long times spent within an application. Familiarity with applications often achieved.	Link to a site, browse or read pages, fill out forms, register for services, participate in transactions, download and save things. Movement between pages and sites very rapid. Familiarity with many sites not established.
User's Conceptual Space	Controlled and constrained by program.	Infinite and generally unorganized.
Presentation Elements	Windows, menus, controls, data, toolbars, messages, and so on. Many transient, dynamically appearing and disappearing. Presented as specified by designer. Generally standardized by toolkits and style guides.	Two components – browser and page. Within page, any combination of text, images, audio, video, and animation. May not be presented as specified by the designer – dependent on browser, monitor, and user specifications. Little standardization.

(continued)



Dimension	Web Application	Web Page	In the middle of continuum
The Nature of the relationship with the user	Users must use the program, Users must identify, themselves or login to be able to do anything, The program must be reliable, and system failure will be immediately noticed Work is remembered from earlier session	The system does not care who the user are, Login is required to simply give access to more information than would be received anonymously, System failure may not be noticed	Minimal information such as a credit card number or address is remembered
The Conversation style	The style is formal	The style is casual , informal and generic	The style is polite but friendly
The Nature of the interaction	A Large amount of data is entered, complex task are being performed, data is being created, manipulated and permanently stored	No data is entered or changed	A small amount of data is entered and possibly stored, Milestones, checkpoints, endpoints are expected and included
The Frequency of use	It is used daily, it is used for long period (4-8 Hour), it is used to help resolve emergencies	It is used only occasionally, it is used for only few minutes at a time, it is used to find out about something	It is used periodically or episodically

The merging of graphical business systems and the web (Cont.)



Dimension	Web Application	Web Page	In the middle of continuum
The Perceived distance of the provider	It is viewed as being local in origin, it is viewed as being controlled locally by a data administrator, the response time is fast	The origin is unknown, the origin is viewed as originating somewhere else in the country or overseas, the response time is slow	The origin cannot be ascertained or doesn't matter
Real Time interaction	Data is fed in real time, the information is critical, a delay is life-threatening, long responses exist but users remain	Time is irrelevant, long response delay exists and causes users to exit	Response is near real time
How much help will users need?	Intense training programs and experience are needed to use and become experts	Every visit is one time session, minimizing or eliminating the need for help	A minimum amount of experience, training or help material is needed
The interaction style	The navigation is controlled, controls are complex, the syntax is object:Action, there is little or no reversibility, drag and drop manipulation is expected, exit requires users to log off	The navigation is flexible and simple, controls are simple, no object selection is required, actions are easily reversed using the back button, no drag-drop manipulation exists, single clicking links are used to navigate	The navigation is flexible, simple data collection exists, user exits by closing web browser
The Presentation style	The style is subdued and serious	The style is colorful, graphic, possibly animated, the controls are obvious and self-explanatory	The style is cooler but attractive
Follow standards	Platform standard compliance exists, the style resembles or matches GUI standard	Few cross-site standards are followed, only intra-site consistency exists	Some common patterns exist

Referensi

- Wilbert O. Galitz, 2007, **The Essential Guide to User Interface Design: An Introduction to GUI Design Principles and Techniques, Third Edition**, Wiley Publishing
- <https://www.youtube.com/watch?v=KtwwustmEDI>
- <https://www.youtube.com/watch?v=gsNQRcSoa9M>